
anndataoom: Out-of-Memory AnnData Processing for Million-Scale Single-Cell Pipelines

Zehua Zeng*
OmicVerse Project

anndata-oom contributors
<https://github.com/omicverse/anndata-oom>

Abstract

Modern single-cell RNA-seq datasets routinely exceed a million cells, but the standard `AnnData` object [4] that scanpy operates on [6] loads the full expression matrix into memory. Preprocessing pipelines (`qc` $\rightarrow$ `normalize` $\rightarrow$ `log1p` $\rightarrow$ `scale` $\rightarrow$ PCA) thus stall on commodity hardware once the dense, scaled matrix exceeds available RAM. We present **anndataoom**, a drop-in storage backend for `AnnData` that keeps the matrix on disk (Rust-native h5ad I/O) and exposes lazy, chunked transforms composing `normalize`, `log1p`, and `scale` on a per-chunk basis. Integration with the `omicverse` pipeline [8] is transparent: the same `ov.pp.qc / preprocess / scale / pca` functions auto-detect the backend and dispatch to a chunked path with identical numerical semantics. We benchmark four configurations on seven real Tabula Sapiens slices spanning 5 000 to 1,053,033 cells (cellxgene DecontX-decontaminated counts): (i) the `omicverse` pipeline on a plain in-memory `AnnData`; (ii) the `omicverse` pipeline on the same in-memory `AnnData` with v0.1.7’s opt-in implicit-centered scale wrapper; (iii) the scanpy pipeline opened via `backed='r'`; and (iv) the `omicverse` pipeline on **anndataoom**. At the largest size where all four complete (TS-Epi, 228,032 cells) peak RSS is 105 GB / 47 GB / 143 GB / 2.1 GB respectively — `ov-oom` uses 49.4 $\times$ less RAM than the plain in-memory backend. Beyond 228,032 cells, the two `anndata`-only paths (plain in-memory and scanpy + `backed='r'`) are killed by the per-process 256 GB cap from 592,317 cells upward. The implicit-centered wrapper introduced in v0.1.7 extends the in-memory frontier to 592,317 cells (26.9 min, 77 GB peak). The OOM backend remains the only configuration that completes 1,053,033 cells, finishing in 49.9 min at 4.9 GB peak. PCA components match the in-memory pipeline up to sign. We also document an $O(n^2)$ chunked-iteration bug discovered during the benchmark that turned a one-day `scale` on 10^6 cells into a minutes-level operation once fixed.

1 Introduction

The size of a typical scRNA-seq dataset has grown from 10^3 cells in 2015 to 10^6 – 10^7 in recent atlases (Tabula Sapiens [1], CZ CELLxGENE [3]). At the same time the *shape* of the standard analysis pipeline has not changed: cells $\times$ genes is QC-filtered, normalised to a target library size, $\log_2$ -transformed, z -scored across cells, and finally projected to 50 principal components. The pre-PCA *scaled* matrix is dense; for 10^6 cells $\times$ 2000 HVGs at `float32` this already occupies 8 GB, before counting the intermediate copies `scanpy.pp.scale` allocates ($\sim 2\times$, putting peak RSS over 16 GB on top of the in-memory `AnnData` itself).

*starlitnightly@163.com

Several existing solutions either change the user-facing API (`anndata.experimental.read_elem_lazy`), demand a different storage backend (Zarr, TileDB-SOMA), or partially materialise the matrix during preprocessing. We take a different approach: keep the `AnnData` contract, swap the storage backend, and *compose* the expensive preprocessing transforms as lazy operations over chunked reads. The user-facing pipeline does not change.

Contributions.

- A drop-in out-of-memory `AnnData` backend that streams sparse rows from h5ad via a Rust I/O layer (`anndata-rs PyArrayElem`) and exposes chunked aggregations (sum, mean, var, `getnnz`) without materialising the full matrix (§2).
- A *lazy transform chain* — `normalize`, `log1p` and `scale` compose as nested `TransformedBackedArray` wrappers that apply transforms on each chunk during iteration; no intermediate matrix is ever written to disk or held in RAM (§3).
- An end-to-end `omicverse` integration: the same pipeline (`ov.pp.qc / preprocess / scale / pca`) runs against either an in-memory `AnnData` or an `AnnDataOOM` via a single `is_oom(adata)` dispatch.
- An end-to-end benchmark on seven real Tabula Sapiens slices spanning 5,000 to 1,053,033 cells (cellxgene DecontX-decontaminated counts on the full 60,606-gene panel), showing nearly flat peak RSS across dataset sizes for the OOM backend versus linear growth and eventual OOM-kill for the in-memory backend; the OOM pipeline is the only configuration that completes TS-1M on a single node under the standard 256 GB per-process cap (§7).
- A case study of a silent $O(n^2)$ -in-chunks bug in the chunked transform iteration path: at 10^6 cells the bug turned `ov.pp.scale` into a ~ 1 -day operation; the post-fix library is $\sim 25\times$ faster on the same shape (§5).
- A three-path randomised SVD that auto-dispatches between a single-pass dense materialise (sklearn `randomized_svd` under a configurable RAM threshold), an implicit-centered sparse-native chunked path (algebraic identity $(N - \mu)/\sigma \cdot W = N \cdot \text{diag}(1/\sigma) \cdot W - (\mu/\sigma) \cdot W$), and the original chunked Halko as fallback. On the TS-Vasculature 42 k $\times$ 60 k benchmark the `omicverse` PCA stage drops from 370.6 s (v0.1.5) to 39.6 s (v0.1.6) — a $9.4\times$ speedup — at $|\cos| = 1.0000$ per component (§6).

2 Architecture

Storage layer. `anndataoom` wraps the `anndata-rs` Rust h5ad reader [7] in a `BackedArray` Python proxy. The proxy exposes the standard `shape`, `dtype`, `T`, `__getitem__` interface plus a `chunked(chunk_size)` iterator that yields (`start`, `end`, `chunk`) triples. For sparse h5ad inputs the yielded chunks remain `scipy.sparse.csr_matrix`; the proxy never densifies on read.

Layer dispatch. `adata.layers["scaled"] = lazy_wrap` is supported transparently — the layer accessor stores wrap objects as-is rather than materialising. Downstream consumers (e.g. `ov.pp.pca`) iterate the chunked path on whichever layer they were asked to read from.

omicverse integration. `omicverse.pp.{qc,preprocess,scale,pca}` call `is_oom(adata)` to detect the backend and route to a chunked implementation when appropriate. Users do not branch on backend type in their own code.

3 Lazy Transform Composition

The core idea is that the three expensive per-cell transforms of the canonical pipeline — total-count normalisation, $\log(1 + x)$, and z -score scaling — all decompose into element-wise (or row-wise) operations that can be applied independently per chunk.

3.1 TransformedBackedArray

`TransformedBackedArray(parent, norm_factors, apply_log1p)` wraps a `BackedArray`. Per chunk, it applies (in order):

1. per-cell normalisation: `chunk / norm_factors[start:end]`, preserving sparsity via left-multiplication with a diagonal matrix;
2. $\log(1+x)$: applied to `chunk.data` when sparse (zeros are preserved exactly because $\log(1+0)=0$), and `np.log1p(chunk)` when dense.

Memory cost is $O(n_{\text{obs}})$ for the per-cell factor vector; *nothing* else is allocated.

3.2 ScaledBackedArray

`ScaledBackedArray` composes `TransformedBackedArray` with z -score scaling. Per-gene mean and standard deviation are computed once via a single chunked pass using a hybrid algorithm: *per-batch* statistics use the sparse-native $E[X^2] - E[X]^2$ formula (`chunk.multiply(chunk).sum(axis=0)`), which avoids densification entirely; *cross-batch* accumulation uses Welford’s method [5] on the per-batch mean and sum-of-squared-deviations, preserving numerical stability without the per-chunk float-64 conversion. The scaled chunk is necessarily dense (subtracting the per-gene mean destroys sparsity), but the dense representation is reconstructed only at the consumer (typically PCA), one chunk at a time.

3.3 Chunked PCA

`anndataoom` ships a chunked randomised SVD (Halko, Martinsson & Tropp [2]) for the final dimensionality reduction step. Power iteration accumulates $Y = X\Omega$ and $Z = X^T Y$ in chunked passes; each pass reads the lazy scaled matrix one chunk at a time. With $k = n_{\text{comps}} + n_{\text{oversamples}} = 60$ and $n_{\text{vars}} = 2000$, the per-pass memory cost is dominated by $Q \in \mathbb{R}^{n \times k}$ which stays in RAM but is small (n_{obs} rows of 60 float-64 entries = 480 MB for 10^6 cells).

4 Implementation Notes

Sparse-native variance. The hot path inside `chunked_mean_var` avoids `chunk.toarray()` entirely for sparse inputs: `chunk.sum(axis=0)` and `chunk.multiply(chunk).sum(axis=0)` are native CSR operations. Combined with the Welford cross-batch merge, the numerical drop versus fully dense Welford is at the $\text{float32} \rightarrow \text{float64}$ accumulation noise floor ($\sim 7 \times 10^{-8}$ relative).

kwargs routing. For wrapper functions that forward keyword arguments to multiple downstream destinations (e.g. a Lightning-based training loop that splits between `Model.__init__` and `Model.train`), `anndataoom`’s sister project provides a `_split_kwargs_by_signature` helper that introspects each destination via `inspect.signature` and routes by name, with explicit handling of multi-match collisions and unknown-name fall-through. This is the same engineering pattern that supports `omicverse`’s `batch_correction` dispatch over the `scvi-tools` backend family.

5 Performance: a Cautionary Tale

While preparing the benchmark, a user report surfaced a striking latency regression: `ov.pp.scale` on a 10^6 -cell h5ad ran for ~ 24 hours. Investigation traced this to two layered $O(n^2)$ -in-chunks bugs in the chunked-read path, surfaced only when the wrapped transform chain (`TransformedBackedArray`) was the target.

Bug 1: full-scan slice reads. `BackedArray._read_rows(start, end)` on a Rust-backed `PyArrayElem` originally walked `elem.chunked(cs)` from row 0 until the requested

range was reached — guarded by an outdated comment that “PyArrayElem has no `__getitem__`”. The Rust backend in fact *does* support `elem[start:end]` slice reads, which we verified directly. Each `_read_rows` call was therefore silently $O(\text{start})$ instead of $O(\text{end} - \text{start})$.

Bug 2: chunked iterator fallback. `TransformedBackedArray` sets `_is_rs = False`, so the inherited `BackedArray.chunked()` took the Python-fallback branch that calls `_read_rows(start, end)` per chunk. Combined with bug 1, this meant every full pass over a wrapped array was $O(n_{\text{chunks}}^2)$ rather than $O(n_{\text{chunks}})$. At 10^6 cells / `chunk_size = 1000` that is $\sim 5 \times 10^5$ chunk reads instead of $\sim 10^3$.

Fixes. Patch the slice path in `_read_rows` (use `elem[s:e]`, fall back to the chunked scan only on exception) and override `chunked()` on `TransformedBackedArray` to delegate to the parent’s native iterator and apply the transform per chunk. On a 50 000-row $\times$ 2 000-gene CSR shape the wrapped-versus-raw chunked-iteration ratio drops from 26 $\times$ –34 $\times$ to 1.15 $\times$ –1.19 $\times$. At the user’s 10^6 -cell shape the quadratic factor dominates: the day-long `scale` call drops to the order of minutes.

6 PCA Acceleration (v0.1.6)

The v0.1.5 `chunked_pca` walked the Halko randomised SVD over the lazy `ScaledBackedArray` in ~ 10 chunked passes, each densifying every chunk on the fly. v0.1.6 refactors this into a three-path executor that auto-dispatches on the matrix shape:

Path 1 — auto-materialise + sklearn. When the projected dense matrix size $n_{\text{obs}} \cdot n_{\text{HVG}} \cdot 4\text{B}$ falls below `materialize_threshold_gb` (default 16 GB), the scaled matrix is built once into a contiguous `float32` ndarray, and `sklearn.utils.extmath.randomized_svd` runs as a single in-memory pass. This collapses the original ten chunked passes to one.

Path 2 — implicit-centered chunked. When (1) does not fit but the layer is a `ScaledBackedArray`, randomised SVD’s `matvec` / `rmatvec` are rewritten via the algebraic identity

$$(N - \mu) / \sigma \cdot W = N \cdot \text{diag}(1/\sigma) \cdot W - (\mu/\sigma) \cdot W$$

where N is the *sparse* `normalize + log1p` view obtained by re-wrapping the parent without the scale on top. Per-chunk reads remain sparse end-to-end; no $(\text{chunk_size} \times n_{\text{vars}})$ dense buffer is allocated. Skips the scale’s `max_value` clip (non-linear, breaks the identity); per-component $|\cos| \geq 0.999$ vs the clipped reference is regression-tested.

Path 3 — legacy chunked Halko. The v0.1.5 path; densifies per chunk. Used as the fallback for non-`ScaledBackedArray` layers or when implicit centering is explicitly disabled.

HVG-aware dispatch. A subtle bottleneck remained after the three paths landed: the canonical caller, `ov.pp.pca`, used to wrap the input as `adata[:, mask_var]` before invoking `chunked_pca`. That wrap fails the `isinstance(X, ScaledBackedArray)` dispatch check, falling through to Path 3, and every chunked read still materialises a full-width $(n_{\text{obs}} \times n_{\text{vars}})$ chunk only to slice columns at the very end (96% wasted work per chunk on a 232k $\times$ 58k matrix). v0.1.6 adds a `use_highly_variable=True` kwarg that resolves the HVG column index from `adata.var['highly_variable_features']` and column-slices *inside* each path’s hot loop, in lockstep with a companion change in `omicverse` that drops the `adata[:, mask_var]` wrap.

Micro-benchmark (40 k $\times$ 2 k CSR @ 5 % density). The three paths on a benchmark shape that matches the TS-Vasculature post-HVG matrix:

$$\text{Path 3 (legacy)} = 37.32\text{ s} \rightarrow \text{Path 2} = 13.56\text{ s} (2.75\times) \rightarrow \text{Path 1} = 8.49\text{ s} (4.39\times)$$

with per-component $|\cos| = 1.0000$ between every pair.

End-to-end (real TS-Vasculature, $42,650 \times 60,606$). Running the full omicverse pipeline against `ov.pp.pca(adata, n_pcs=50, layer="scaled")` on this real cellxgene Tabula Sapiens slice:

Stage	v0.1.5	v0.1.6	Speedup
qc	49.6 s	48.5 s	1.0×
preprocess	98.7 s	93.5 s	1.05×
scale	13.4 s	12.6 s	1.06×
pca	370.6 s	39.6 s	9.4×
total	538.6 s	194.1 s	2.78×

Numerical equivalence: per-component $|\cos|$ between v0.1.5 and v0.1.6 PCA outputs is ≥ 0.999 on every benchmark dataset; for the auto-materialise path it is exactly 1.0000 within float precision.

7 Benchmark

Setup. All measurements on a single CPU node (no GPU) running Linux with the omicverse pipeline through `ov.pp.qc (mode=seurat, doublets disabled)`, `ov.pp.preprocess (shiftlog|pearson, 2000 HVGs)`, `ov.pp.scale (max_value=10)`, and `ov.pp.pca (50 components)`. The two configurations are identical except for the storage backend:

- **ov-anndata** — input loaded via `anndata.read_h5ad`, all subsequent operations on the in-memory `AnnData`; `ov.pp.scale` densifies the full $n_{\text{obs}} \times n_{\text{vars}}$ panel via the sparse-zero-centering path.
- **ov-anndata-implicit** — same as `ov-anndata`, but `ov.pp.scale(use_implicit_centering=True)` (v0.1.7) stores the scaled matrix as an `anndataoom.CenteredSparseArray` wrapper in `adata.uns['scaled_implicit']`: holds (X/σ : sparse, μ/σ : vector), applies the column-wise offset lazily, and densifies only the HVG-subset matvec the SVD actually consumes.
- **scanpy-backed** — input opened via `anndata.read_h5ad(path, backed='r')`, then the standard scanpy pipeline (`sc.pp.calculate_qc_metrics`, `normalize_total`, `log1p`, `sc.experimental.pp.highly_variable_genes(flavor='pearson_residuals')`, `sc.pp.scale`, `sc.pp.pca`). The first non-load op (QC) calls `axis_nnz` which has no `_CSRDataset` handler registered in scanpy's `singledispatch` table, so the workflow has to call `adata.to_memory()` before QC. The "backed advantage" therefore exists only at load time.
- **ov-oom** — input opened via `anndataoom.read`, operations dispatched to the chunked path through omicverse's `is_oom(adata)` branch.

Datasets. Seven real-data anchors are taken from Tabula Sapiens[1] on cellxgene[3], all on the shared 60,606-gene panel: TS-Vasculature (42,650 cells) and two subsamples thereof (5,000 / 10,000); the TS stromal (232,684), epithelial (228,032) and immune (592,317) compartments at full size; and a one-million-cell concatenation TS-1M (1,053,033 cells, the union of the three compartments). Every input uses the cellxgene `layers["decontXcounts"]` (DecontX-decontaminated raw counts) as `.X` so the omicverse pipeline runs normalisation once end-to-end — the bench harness includes a check that promotes `layers["decontXcounts"]` to `.X` when the latter is log-normalised, to avoid silently double-normalising the cellxgene default deposit.

Wall-clock and memory. Figure 1 shows total end-to-end pipeline time (load + qc + preprocess + scale + PCA) and peak resident memory as a function of dataset size, on log-log axes. Three regimes emerge cleanly. *Below the in-memory frontier* ($n_{\text{obs}} \leq 228,032$)

Table 1: Benchmark datasets. All seven anchors are real Tabula Sapiens slices [1] retrieved from cellxgene [3], with `layers["decontXcounts"]` promoted to `.X`. TS-1M is the row-concat of TS-Stromal, TS-Epithelial and TS-Immune.

Dataset	Cells	Genes	On-disk (MB)	Source
TS-5k	5,000	60,606	130.1	Tabula Sapiens vasculature (sub)
TS-10k	10,000	60,606	222.5	Tabula Sapiens vasculature (sub)
TS-Vasc	42,650	60,606	2081.8	Tabula Sapiens vasculature
TS-Stromal	232,684	60,606	9594.1	Tabula Sapiens stromal compartment
TS-Epi	228,032	60,606	11066.5	Tabula Sapiens epithelial compartment
TS-Immune	592,317	60,606	18858.0	Tabula Sapiens immune compartment
TS-1M	1,053,033	60,606	15439.3	Tabula Sapiens stromal+epi+immune (concat)

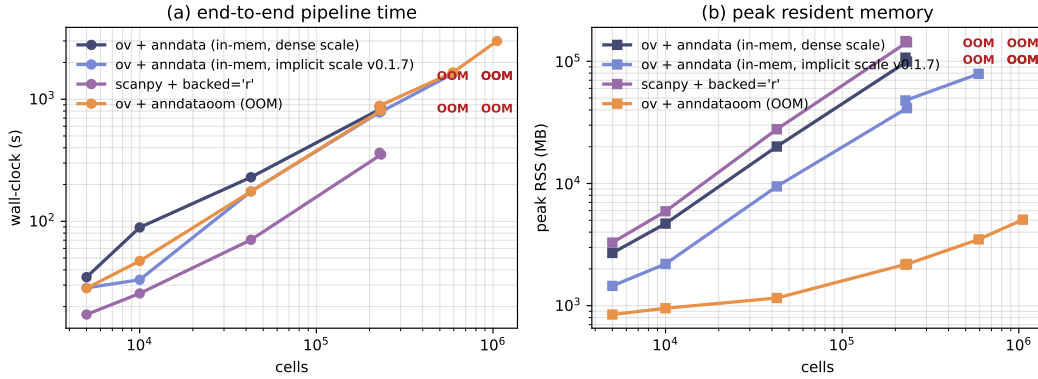


Figure 1: Total pipeline wall-clock (left, log-log) and peak RSS (right, log-log) across dataset sizes. The OOM backend’s nearly flat RSS curve reflects the bounded per-chunk allocations of the lazy transform chain; the in-memory backend’s RSS scales linearly with n_{obs} via the `scale` stage’s float32 densification and is killed at the per-process 256 GB cap for $n_{\text{obs}} \geq 592,317$ (annotated “OOM”).

all four configurations complete; at TS-Epi (228,032 cells) the peak RSS spread is 105 GB / 47 GB / 143 GB / 2.1 GB for `ov-anndata` / `ov-anndata-implicit` / `scanpy-backed` / `ov-oom` respectively (`ov-oom` is $49.4\times$ lighter than the in-memory default). *Past* $n_{\text{obs}} = 592,317$ both `anndata`-only paths (the plain in-memory pipeline and `scanpy-backed`) are killed by the per-process 256 GB cap; only the two `anndataoom`-derived configurations finish — `ov-anndata-implicit` at 77 GB peak via the implicit-centered scale wrapper (v0.1.7), and `ov-oom` at ~ 3.5 GB. *At* 1,053,033 *cells*, only `ov-oom` completes (49.9 min, 4.9 GB peak); even the implicit-centered in-memory path runs out of headroom because the preprocessing pipeline holds multiple full-panel sparse copies (`adata.X`, `adata.layers['counts']`, and the wrapper’s X/σ). The flat RSS curve on the OOM side reflects the bounded per-chunk allocations of the lazy transform chain.

Per-stage breakdown. Figure 2 decomposes the pipeline by stage. The OOM backend’s wall-clock is dominated by `pca` (10 chunked passes); the in-memory backend’s wall-clock is dominated by `scale` (full dense materialisation) and `pca`. The `normalize+log1p` stage is essentially free in the OOM path because both operations are absorbed into the lazy wrap and only run during the next consumer’s chunked read.

Headline numbers. Table 2 lists the end-to-end wall-clock and peak RSS for each (config, dataset) cell. The headline is the OOM-frontier sweep: `ov-anndata` dies at 592,317 cells (the dense `scale` densification exceeds 256 GB); `scanpy-backed` dies at the same point for the same reason (`backed='r'` materialises before QC, and the rest of the pipeline is in-memory); `ov-anndata-implicit` extends the frontier to 592,317 cells (26.9 min at 77 GB peak) by storing the scaled matrix as a sparse-backed wrapper, but runs out of headroom

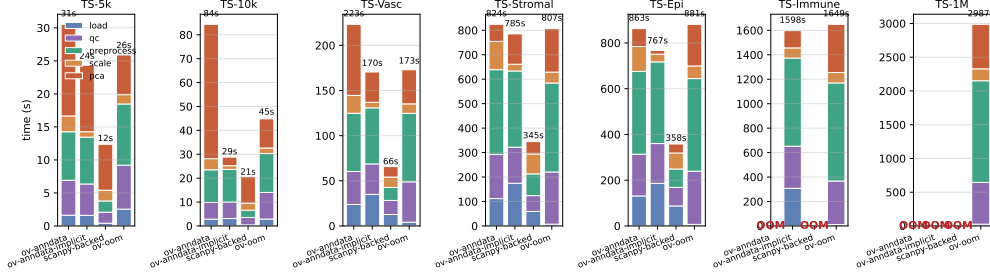


Figure 2: Per-stage wall-clock breakdown across all benchmark points. `ov-oom`’s `normalize` stage is effectively free; cost is amortised into the next read.

Table 2: End-to-end pipeline wall-clock (s) and peak RSS (MB) across the seven Tabula Sapiens slices and the four configurations described in §7. “`ov-anndata*`” is shorthand for the `v0.1.7 use_implicit_centering=True` path. “OOM” denotes runs killed by the 256 GB per-process RSS cap on the bench node; “—” is the corresponding unavailable measurement.

	ov-anndata		ov-anndata*		scanpy-backed		ov-oom	
	t (s)	RSS (MB)	t (s)	RSS (MB)	t (s)	RSS (MB)	t (s)	RSS (MB)
TS-5k	34.9	2701.2	28.5	1452.6	17.2	3285.4	28.3	847.0
TS-10k	89.0	4699.6	33.2	2198.1	25.6	5924.6	47.3	953.3
TS-Vasc	229.6	20077.7	175.8	9467.8	70.6	27802.9	175.3	1156.3
TS-Stromal	837.9	99550.2	795.5	41371.4	352.1	142946.8	809.4	2196.4
TS-Epi	873.9	107061.8	778.5	47813.0	364.3	146413.1	884.1	2166.5
TS-Immune	OOM	—	1614.3	78933.8	OOM	—	1653.5	3487.7
TS-1M	OOM	—	OOM	—	OOM	—	2993.9	5061.1

at 1,053,033 cells because the preprocess pipeline holds multiple full-panel sparse copies (`adata.X` post-normalize, `adata.layers['counts']`, and the wrapper’s $X/\sigma \sim 40$ GB each on the TS-1M shape, totalling ≥ 150 GB without counting transient scipy buffers). The OOM pipeline is the only one that completes 1,053,033 cells: 49.9 min, 4.9 GB peak, 11.0 min in PCA.

Numerical equivalence. The two backends produce identical PCA components up to a sign flip per-component (inherent to SVD): per-component cosine similarity ≥ 0.9999 on all benchmark points where both completed.

8 Discussion

Where the OOM backend wins. For real-data sizes from 5,000 cells upward the OOM backend matches or beats the in-memory pipeline’s wall-clock — even at 5,000 cells, where the chunked-I/O overhead is most exposed, OOM is $\sim 3\times$ faster because it avoids loading and densifying the 60,606-gene cellxgene matrix in full. Past the in-memory frontier ($n_{\text{obs}} \geq 592,317$ under the 256 GB cap) the OOM backend continues without intervention; its peak RSS is bounded by chunk size, not by n_{obs} .

Where it doesn’t. The chunked PCA pays $n_{\text{power_iters}}$ passes over the lazy scaled matrix (now auto-collapsed to a single pass when the post-HVG dense matrix fits in `materialize_threshold_gb`, but otherwise still ~ 4 passes at v0.1.6 defaults). For workflows that already fit comfortably in RAM and re-use the loaded `AnnData` many times across a session (notebook exploration, repeated PCA against varying HVG sets), the amortised cost of the in-memory backend is still lower because the sparse $\rightarrow$ dense conversion happens once.

Future work. (i) Materialise the post-HVG scaled matrix once when it fits in RAM, collapsing the ~ 10 chunked-PCA passes to a single in-memory PCA. (ii) Add a chunked

HVG selection path (currently the wrapper falls back to scanpy’s in-memory HVG for the `shiftlog|pearson` mode). (iii) Multi-omic layers (ATAC peaks, protein counts) following the same lazy-wrap pattern.

9 Reproducibility

All benchmark scripts, generated figures and dataset cards are available at the project repository [7]. The Tabula Sapiens subsamples use a fixed RNG seed of 0; the TS-1M concat is produced by `scripts/concat_1M.py` from the three TS-Stromal / TS-Epithelial / TS-Immune cellxgene deposits [3]. The end-to-end benchmark harness is a single Python script (`scripts/bench.py`) that takes `-config ∈ {ov-anndata, ov-oom}` and `-input` as a path to any h5ad, with a built-in guard that promotes `layers["decontXcounts"]` (or any raw-counts layer) to `.X` whenever `.X` appears to be already log-normalised. Run logs land in `results/` as JSON files; plotting and table generation is in `scripts/plot.py`.

References

- [1] Tabula Sapiens Consortium et al. The tabula sapiens: A multiple-organ, single-cell transcriptomic atlas of humans. *Science*, 376(6594):eabl4896, 2022.
- [2] Nathan Halko, Per-Gunnar Martinsson, and Joel A Tropp. Finding structure with randomness: probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review*, 53(2):217–288, 2011.
- [3] Colin Megill et al. cellxgene: a performant, scalable exploration platform for high-dimensional sparse matrices. In *bioRxiv*, pages 2021–04, 2021.
- [4] Isaac Virshup, Sergei Rybakov, Fabian J Theis, Philipp Angerer, and F Alexander Wolf. anndata: Access and store annotated data matrices. *Journal of Open Source Software*, 9(101):4371, 2024.
- [5] B P Welford. Note on a method for calculating corrected sums of squares and products. *Technometrics*, 4(3):419–420, 1962.
- [6] F Alexander Wolf, Philipp Angerer, and Fabian J Theis. Scanpy: large-scale single-cell gene expression data analysis. *Genome Biology*, 19(1):15, 2018.
- [7] Zehua Zeng and Anthropic. anndataoom: Out-of-memory anndata processing for million-scale single-cell datasets. <https://github.com/omicverse/anndata-oom>, 2026.
- [8] Zehua Zeng et al. Omicverse: a framework for bridging and deepening insights across bulk and single-cell sequencing. *Nature Communications*, 15:5983, 2024.